SOFTWARE REQUIREMENTS AND DESIGN SPECIFICATION

Present Call

AI-Based Attendance System using Face and Voice Biometrics

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1. Introduction

1.1 Purpose

This document is the Software Requirements and Design Specification (SRDS) for Present Call, an AI-based attendance management system that identifies students using face and voice biometrics. It defines what the system must do, how it is designed and built, how it is tested and deployed, and how it will be maintained and extended. It is intended to serve as the single reference for development, quality assurance, deployment, academic evaluation and future enhancement of the product.

1.2 Scope

Present Call allows a teacher to mark the attendance of a whole class from one or more group photographs, or through a sequential voice roll-call, and gives students a personal dashboard of their attendance. The scope of this specification is summarised below.

Table 1: Scope of the specification

Area	Description
In scope	Teacher and student web portals; user registration and login; subject (course) management; QR / link based enrolment; face and voice biometric registration; multi-face attendance from photographs; voice-based attendance; liveness / anti-spoofing checks; review and confirmation of results; attendance records and CSV export; student dashboard; cloud database; Flask landing page; deployment on cloud platforms.
Out of scope (v1.0)	Integration with institutional ERP / LMS; payment or billing; native mobile applications; offline operation; payroll or workplace time-tracking; guardian SMS / e-mail delivery (planned for a later release); dedicated institution-administrator console.

1.3 Intended Audience

Table 2: Intended audience

Audience	How they use this document
Project guide / examiners	Evaluate completeness of requirements, design quality, testing and engineering practice.
Developers	Implement and extend features; understand architecture, data model and AI pipelines.
QA engineers	Derive test cases from requirements; use evaluation methodology for biometric accuracy.
DevOps / maintainers	Deploy, configure, monitor, back-up and troubleshoot the system.
Institution stakeholders	Assess suitability, security, privacy implications and roadmap.

1.4 Document Conventions

Requirements carry unique identifiers: **FR-<MODULE>-<nn>** for functional requirements and **NFR-<CATEGORY>-<nn>** for non-functional requirements. Priority follows the MoSCoW scale (Must, Should, Could). The **Status** column shows whether a feature is **Implemented** in the current release or **Planned** for a later release. Quantitative non-functional values are **targets** unless explicitly described as observed results.

1.5 Definitions, Acronyms and Abbreviations

Table 3: Definitions and abbreviations

Term	Meaning
Biometric template	Numerical vector (embedding) derived from a person's face or voice and used for matching; treated as sensitive personal data.
Embedding	Fixed-length vector produced by a neural network; 128-d for faces (dlib ResNet) and 256-d for voice (Resemblyzer).
HOG	Histogram of Oriented Gradients, the feature type used by dlib's frontal face detector.
SVC	Support Vector Classifier (scikit-learn, linear kernel) used to classify face embeddings.
L2 distance	Euclidean distance between two embeddings; smaller means more similar.
Cosine similarity	Similarity measure between two voice embeddings; higher means more similar.
Liveness / anti-spoofing	Checks that reject printed photographs, screen replays and recorded voices.
FAR / FRR / EER	False Acceptance Rate, False Rejection Rate, Equal Error Rate.
RLS	Row Level Security, a PostgreSQL feature restricting which rows a user can access.
PostgREST	Service that exposes a PostgreSQL database as a REST API (used by Supabase).
DPDP Act	Digital Personal Data Protection Act, 2023 (India).
SRDS	Software Requirements and Design Specification (this document).

1.6 References

1. IEEE Std 29148-2018, Systems and software engineering — Life cycle processes — Requirements engineering.
2. IEEE Std 1016-2009, Standard for Information Technology — Systems Design — Software Design Descriptions.
3. ISO/IEC 25010:2011, Systems and software quality models.
4. ISO/IEC/IEEE 29119, Software testing standards.
5. OWASP Top 10 and OWASP Application Security Verification Standard (ASVS).
6. Digital Personal Data Protection Act, 2023 (Government of India).
7. D. E. King, "Dlib-ml: A machine learning toolkit," JMLR, vol. 10, 2009.
8. F. Schroff, D. Kalenichenko, J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," CVPR, 2015.
9. L. Wan, Q. Wang, A. Papir, I. L. Moreno, "Generalized end-to-end loss for speaker verification," ICASSP, 2018.
10. Official documentation: Streamlit, Supabase, scikit-learn, librosa, Resemblyzer, Flask.

2. Product Overview

2.1 Background and Business Need

Institutions require accurate attendance records because attendance often determines examination eligibility and compliance reporting. Manual roll-call consumes lecture time, and paper registers are prone to errors and proxy marking, where one student answers for another. Single-factor technologies

such as RFID cards or face-only recognition can be bypassed with a borrowed card or a printed photograph. Present Call addresses this need with two independent biometric factors, group-photo processing for speed, and cloud dashboards for transparency.

2.2 Business Objectives and Success Criteria

Table 4: Business objectives

ID	Objective	Measure	Target
BO-1	Reduce time spent on attendance	Time to mark attendance for one class	Under 1 minute per class
BO-2	Reduce proxy attendance	Rejection of spoof attempts in testing	All tested photo / replay spoofs rejected
BO-3	Accurate identification	Identification accuracy	≥ 95% (observed: 95.8%)
BO-4	Transparent records	Student access to own attendance	Real-time dashboard for every student
BO-5	Low cost of ownership	Infrastructure and licence cost	Open-source stack; free-tier hosting

2.3 Product Perspective

Present Call is a self-contained, cloud-hosted web application. Users access it through a browser on a laptop or smartphone with a camera and microphone. The application interacts with Supabase for storage, is deployed from GitHub to Streamlit Community Cloud, and is introduced to visitors through a separate Flask landing page hosted on Vercel. No installation is required on end-user devices.

2.4 Product Features at a Glance

Table 5: Feature summary

Feature	Summary
Multi-face photo attendance	Identify every enrolled student visible in one or more classroom photos.
Voice roll-call	Students say “present” in turn; speaker embeddings identify who spoke.
Anti-spoofing	Model-based liveness check against printed photos, screen replay and recorded voice.
QR / link enrolment	Students join a subject by scanning a QR code or entering a subject code.
Review before save	Teacher reviews the generated report and confirms or discards it.
Records and export	Session history with present / total counts and CSV export.
Student dashboard	Per-subject total classes and classes attended.
Themed responsive UI	Light / dark theme, keyboard shortcuts, responsive layout.

2.5 User Classes and Characteristics

Table 6: User classes

User class	Characteristics	Primary needs
Teacher	Faculty member; moderate technical skill; uses laptop in classroom.	Fast, reliable attendance; correctable results; records and export.
Student	Enrolled learner; uses phone or laptop; basic technical skill.	Simple enrolment, one-time biometric registration, visibility of own attendance.
Institution administrator (planned)	Department head or academic office.	Cross-course reports, policy settings, guardian alerts.
Maintainer / developer	Technical team.	Clear architecture, configuration, logs and deployment procedure.

2.6 Operating Environment

- Client: current versions of Chrome, Edge, Firefox or Safari on Windows, macOS, Linux, Android or iOS, with camera and microphone permission.
- Server: Python 3 runtime on Streamlit Community Cloud with dlib, scikit-learn, Resemblyzer and librosa installed.
- Database: Supabase-hosted PostgreSQL accessed over HTTPS through the supabase-py client.
- Landing page: Flask application on Vercel.

2.7 Design and Implementation Constraints

- The application layer is written in Python using Streamlit; navigation is implemented with st.session_state rather than Streamlit's native multipage mechanism.
- Face recognition uses pre-trained dlib models and an SVC trained on enrolled students' embeddings; no custom large-scale training dataset is used.
- Free-tier hosting imposes limits on memory, CPU, concurrency and idle behaviour, which constrain batch size and latency.
- Biometric data is sensitive personal data and must be handled in line with applicable law (see Section 11).

2.8 Assumptions and Dependencies

- Students are enrolled once with a clear frontal face image and a clean voice sample.
- Classroom photographs are taken with adequate lighting, and faces are reasonably frontal and unobstructed.
- Teachers and students have internet access during attendance sessions.
- Third-party services (Streamlit Community Cloud, Supabase, Vercel, GitHub) remain available and compatible.
- Institutions obtain consent from students before collecting biometric data.

3. Use Cases

3.1 Use Case Diagram

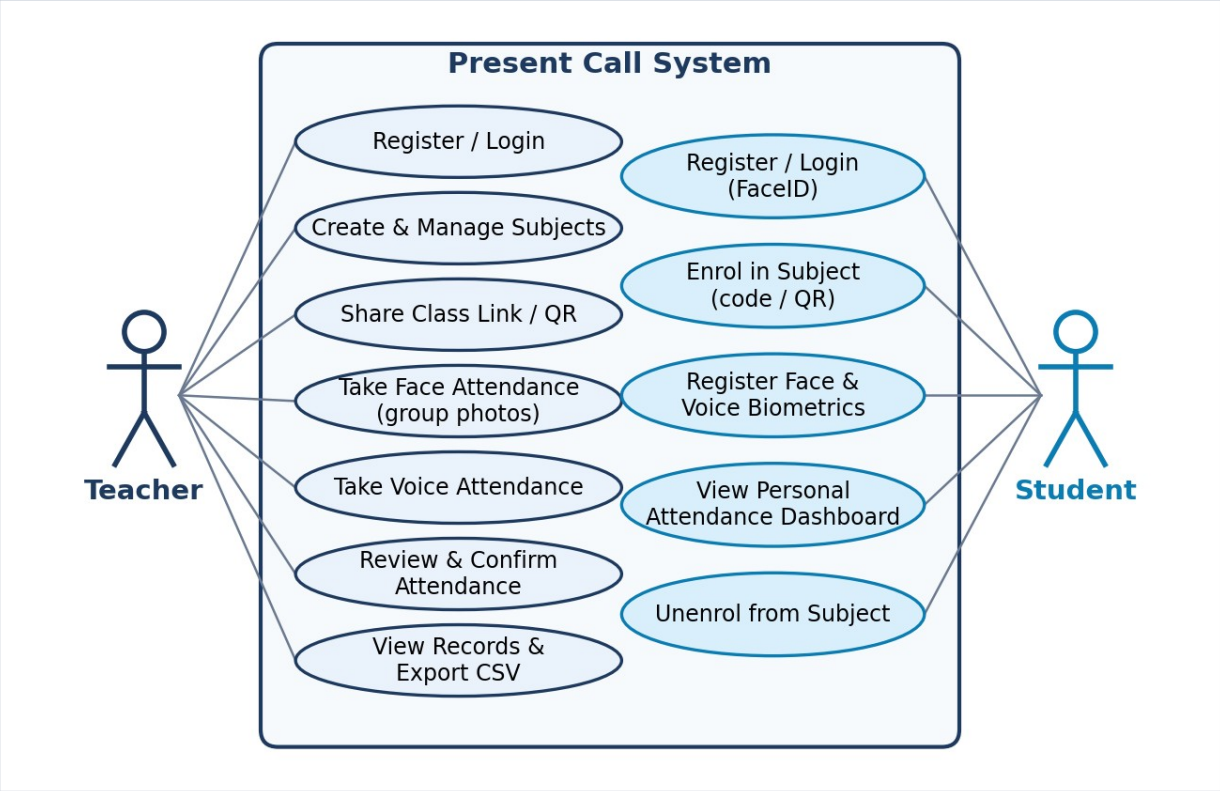


Figure 1: Use case diagram of Present Call

3.2 Use Case Summary

Table 7: Use case summary

ID	Use case	Actor	Precondition	Main flow	Outcome
UC-01	Register / log in	Teacher	Account not logged in	Enter username and password; submit (Ctrl+Enter); or choose Register instead.	Teacher dashboard opens
UC-02	Create subject	Teacher	Teacher logged in	Open Manage Subjects; enter subject code, name, section; create.	Subject card appears
UC-03	Share class link / QR	Teacher	Subject exists	Click Share Code; copy link or show QR code.	Students can join
UC-04	Register and log in with FaceID	Student	Webcam available	Capture face photo; system computes embedding; login by FaceID thereafter.	Student dashboard opens
UC-05	Enrol in subject	Student	Student logged in	Click Enroll in Subject; enter subject code (or scan QR).	Subject added to dashboard
UC-06	Take face attendance	Teacher	Students enrolled with face	Select subject; add photos; Run Face Analysis.	Attendance report generated

ID	Use case	Actor	Precondition	Main flow	Outcome
UC-07	Take voice attendance	Teacher	Students enrolled with voice	Select subject; Use Voice Attendance; record audio; Analyze Audio.	Attendance report generated
UC-08	Review and confirm	Teacher	Report generated	Inspect name, ID, source, status; Confirm & Save or Discard.	Session saved or dropped
UC-09	View records / export	Teacher	Sessions saved	Open Attendance Records; view sessions; export CSV.	Records displayed / file downloaded
UC-10	View dashboard	Student	Student enrolled	Open dashboard; view total and attended classes per subject.	Attendance visible
UC-11	Unenrol	Student	Enrolled in subject	Click Unenroll from this course.	Subject removed

3.3 Detailed Use Case: UC-06 Take Face Attendance

Actor	Teacher (primary); AI engine and database (supporting)
Trigger	Teacher opens Take Attendance and selects a subject.
Preconditions	Teacher is authenticated; subject has enrolled students with registered face embeddings; at least one class photo is available.
Main flow	1. Teacher selects the subject. 2. Teacher adds one or more photographs (Photo 1…n). 3. Teacher clicks Run Face Analysis. 4. System detects and aligns all faces, extracts 128-d embeddings and performs the liveness check. 5. System classifies each face with the SVC and validates it with the L2 distance threshold. 6. System builds the report: each enrolled student is Present (with source photo) or Absent. 7. Teacher reviews the report (UC-08).
Alternative flows	A1. No face detected in a photo: the photo is reported as containing no faces; the teacher may add another photo. A2. Unknown face: not matched to any student and ignored in the report. A3. Teacher clicks Clear all photos: the photo list is emptied.
Exceptions	E1. No students enrolled: the system informs the teacher that no roster exists. E2. Model or database failure: an error message is shown and nothing is saved.
Postcondition	A pending attendance report is available for confirmation; nothing is persisted until the teacher confirms.

3.4 Detailed Use Case: UC-07 Take Voice Attendance

Actor	Teacher (primary); AI engine and database (supporting)
Trigger	Teacher clicks Use Voice Attendance for the selected subject.

Preconditions	Teacher authenticated; microphone permission granted; enrolled students have registered voice embeddings.
Main flow	1. Teacher starts recording the classroom audio. 2. Students say “present” one after another. 3. Teacher clicks Analyze Audio. 4. System resamples the audio to 16 kHz and splits it into speech segments by silence detection. 5. System extracts a 256-d embedding for each segment and compares it with stored voice embeddings using cosine similarity. 6. Matching students are marked Present; others remain Absent. 7. Teacher reviews the report (UC-08).
Alternative flows	A1. Overlapping speech or noise: unmatched segments are ignored and the teacher may re-record.
Exceptions	E1. Microphone permission denied: the system displays guidance to enable it. E2. Silence-only recording: no segments found; the teacher is asked to record again.
Postcondition	A pending attendance report is available for confirmation.

4. Functional Requirements

4.1 Authentication and Account Management

Table 8: Authentication requirements

ID	Requirement	Priority	Status
FR-AUT-01	The system shall allow a teacher to register with a unique username and password.	Must	Implemented
FR-AUT-02	The system shall store passwords only as bcrypt hashes and never in plain text.	Must	Implemented
FR-AUT-03	The system shall authenticate teachers by username and password and support logout.	Must	Implemented
FR-AUT-04	The system shall allow a student to register and log in using FaceID captured through the webcam.	Must	Implemented
FR-AUT-05	The system shall maintain the logged-in role and current page in session state and show role-appropriate pages only.	Must	Implemented
FR-AUT-06	The system shall provide keyboard shortcuts (Ctrl+Enter to log in, Ctrl+Backspace to log out).	Could	Implemented
FR-AUT-07	The system shall limit repeated failed login attempts (rate limiting / lock-out).	Should	Planned
FR-AUT-08	The system shall provide password reset for teacher accounts.	Could	Planned

4.2 Subject Management

Table 9: Subject management requirements

ID	Requirement	Priority	Status
FR-SUB-01	A teacher shall be able to create a subject by entering a unique subject code, subject name and section.	Must	Implemented
FR-SUB-02	The system shall list the teacher's subjects with code, section, number of enrolled students and number of classes held.	Must	Implemented
FR-SUB-03	The system shall generate a shareable class link and a QR code for each subject.	Must	Implemented
FR-SUB-04	A teacher shall be able to edit or delete a subject.	Could	Planned

4.3 Enrolment

Table 10: Enrolment requirements

ID	Requirement	Priority	Status
FR-ENR-01	A student shall be able to enrol in a subject by entering the subject code or by scanning the QR code.	Must	Implemented
FR-ENR-02	The system shall prevent duplicate enrolment of the same student in the same subject.	Must	Implemented
FR-ENR-03	A student shall be able to unenrol from a subject.	Should	Implemented
FR-ENR-04	The system shall update the enrolled-student count on the teacher's subject card.	Should	Implemented

4.4 Biometric Registration

Table 11: Biometric registration requirements

ID	Requirement	Priority	Status
FR-BIO-01	The system shall capture a student's face through the webcam and compute a 128-dimensional embedding.	Must	Implemented
FR-BIO-02	The system shall capture a student's voice sample and compute a 256-dimensional embedding.	Must	Implemented
FR-BIO-03	The system shall reject a registration image in which no face, or more than one face, is detected.	Should	Planned
FR-BIO-04	The system shall store derived embeddings for matching and shall not retain raw biometric images or audio after processing.	Must	Planned

4.5 Face Attendance

Table 12: Face attendance requirements

ID	Requirement	Priority	Status
FR-FAC-01	The teacher shall be able to select a subject and add multiple class photographs.	Must	Implemented

ID	Requirement	Priority	Status
FR-FAC-02	The system shall detect all faces in each photograph using the HOG-based detector.	Must	Implemented
FR-FAC-03	The system shall align faces with 68-point landmarks and generate 128-d embeddings.	Must	Implemented
FR-FAC-04	The system shall identify each face using an SVC (linear kernel) and validate it with an L2 distance check against a threshold (default 0.6).	Must	Implemented
FR-FAC-05	The system shall record which photo identified each student.	Should	Implemented
FR-FAC-06	The system shall perform a liveness / anti-spoofing check and reject printed or screen-displayed faces.	Must	Implemented
FR-FAC-07	The teacher shall be able to clear all added photos.	Should	Implemented
FR-FAC-08	The system shall treat unmatched faces as unknown and not mark any student present for them.	Must	Implemented

4.6 Voice Attendance

Table 13: Voice attendance requirements

ID	Requirement	Priority	Status
FR-VOI-01	The teacher shall be able to record classroom audio in the browser.	Must	Implemented
FR-VOI-02	The system shall resample audio to 16 kHz and split it into speech segments using silence detection.	Must	Implemented
FR-VOI-03	The system shall generate a 256-d embedding per segment and compare it with stored embeddings using cosine similarity.	Must	Implemented
FR-VOI-04	The system shall mark a student present only when the best match meets the similarity threshold.	Must	Implemented
FR-VOI-05	The system shall reject replayed or synthetic recordings through the anti-spoofing model.	Should	Implemented

4.7 Review and Confirmation

Table 14: Review and confirmation requirements

ID	Requirement	Priority	Status
FR-REV-01	The system shall present an attendance report listing student name, ID, source and status before saving.	Must	Implemented
FR-REV-02	The teacher shall be able to Confirm & Save the report, persisting it to the database.	Must	Implemented
FR-REV-03	The teacher shall be able to Discard the report without persisting any data.	Must	Implemented
FR-REV-04	The teacher shall be able to manually override a student's status before saving.	Could	Planned

4.8 Records and Reports

Table 15: Records and reporting requirements

ID	Requirement	Priority	Status
FR-REC-01	The system shall list saved sessions with time, subject, subject code and present / total statistics.	Must	Implemented
FR-REC-02	The system shall allow export of attendance data as CSV.	Should	Implemented
FR-REC-03	The system shall display attendance trend charts.	Could	Implemented
FR-REC-04	The system shall export audit-ready PDF reports with confidence scores and timestamps.	Could	Planned

4.9 Student Dashboard

Table 16: Student dashboard requirements

ID	Requirement	Priority	Status
FR-DSH-01	The system shall show each enrolled subject with code, section, total classes and classes attended.	Must	Implemented
FR-DSH-02	The system shall display the attendance percentage per subject.	Should	Planned
FR-DSH-03	The system shall warn a student when attendance approaches the configured minimum criterion.	Should	Planned

4.10 Administration and Settings

Table 17: Administration requirements

ID	Requirement	Priority	Status
FR-ADM-01	The system shall allow an authorised user to adjust the face similarity threshold.	Could	Planned
FR-ADM-02	The system shall send SMS / e-mail alerts to guardians on absence.	Could	Planned
FR-ADM-03	The system shall provide an institution-administrator role with cross-course reports.	Could	Planned
FR-ADM-04	The system shall support light and dark themes selectable by the user.	Should	Implemented

> **Note:** Status reflects the confirmed implementation summary of the project team and should be re-verified at every release.

5. Non-Functional Requirements

Non-functional requirements follow the ISO/IEC 25010 quality characteristics. Values are design targets to be verified during testing, except the accuracy value that is explicitly marked as observed.

5.1 Performance and Accuracy

Table 18: Performance and accuracy requirements

ID	Requirement	Measure / target
NFR-PER-01	Face analysis of a classroom photograph shall complete within an acceptable interactive time.	≤ 10 s for a photo with up to 30 faces (target)
NFR-PER-02	Voice analysis of a classroom recording shall complete within an acceptable time.	≤ 15 s for a 2-minute recording (target)
NFR-PER-03	Identification accuracy shall meet the project objective.	≥ 95% (observed: 95.8% in internal testing)
NFR-PER-04	False acceptance of impostors shall be minimal.	FAR ≤ 1% at the operating threshold (target; to be validated)
NFR-PER-05	Dashboards and record pages shall load quickly.	≤ 3 s on a normal broadband connection (target)

5.2 Security

Table 19: Security requirements

ID	Requirement	Measure / target
NFR-SEC-01	All traffic between browser, application and database shall use HTTPS / TLS.	100% of endpoints
NFR-SEC-02	Passwords shall be hashed with bcrypt with per-user salt.	No plain-text storage
NFR-SEC-03	Database access shall be restricted by Row Level Security policies in production.	Least-privilege policies (see Section 11.6)
NFR-SEC-04	Secrets (database URL and keys) shall be kept outside source control.	Platform secrets store
NFR-SEC-05	Users shall only access data they are authorised to see.	No cross-user data exposure
NFR-SEC-06	The system shall resist presentation attacks using photos, screens and recordings.	All tested spoofs rejected (target)

5.3 Privacy and Compliance

Table 20: Privacy and compliance requirements

ID	Requirement	Measure / target
NFR-PRV-01	Biometric data shall be collected only with informed consent for the stated purpose of attendance.	Consent captured at registration
NFR-PRV-02	Only derived embeddings shall be stored; raw images and audio shall not be retained after processing.	Verified in design review
NFR-PRV-03	Users shall be able to request correction or deletion of their biometric data.	Deletion within a defined period
NFR-PRV-04	The system shall follow data-minimisation and retention limits aligned with the DPDP Act, 2023.	Documented retention policy

5.4 Reliability and Availability

Table 21: Reliability requirements

ID	Requirement	Measure / target
NFR-REL-01	The service shall be available during working hours.	≥ 99% monthly (best-effort on free-tier hosting)
NFR-REL-02	A failed save shall never produce a partially saved attendance session.	Atomic insert of a session
NFR-REL-03	Data shall be recoverable after failure.	RPO ≤ 24 h, RTO ≤ 4 h (target)

5.5 Usability, Compatibility and Accessibility

Table 22: Usability and compatibility requirements

ID	Requirement	Measure / target
NFR-USA-01	A teacher shall be able to complete an attendance session without training.	≤ 5 steps from login to saved report
NFR-USA-02	The interface shall be responsive for desktop and mobile screens.	360 px to 1920 px width
NFR-USA-03	The interface shall maintain readable contrast in both light and dark themes.	WCAG 2.1 AA contrast (target)
NFR-CMP-01	The system shall work on current browsers.	Chrome, Edge, Firefox, Safari (latest two versions)

5.6 Scalability, Maintainability and Portability

Table 23: Scalability, maintainability and portability requirements

ID	Requirement	Measure / target
NFR-SCA-01	The system shall support typical class sizes and a campus-level user base.	120 students per class; 5,000 users (target)
NFR-MNT-01	Code shall be modular with separate modules for UI, AI, data access and utilities.	Documented module boundaries
NFR-MNT-02	Core logic shall be covered by automated tests.	≥ 70% coverage of non-UI modules (target)
NFR-PRT-01	The application shall be deployable on any host supporting Python 3 and the listed dependencies.	Reproducible from requirements file
NFR-OBS-01	Errors and key events shall be logged without recording biometric data.	Structured logs

6. External Interface Requirements

6.1 User Interfaces

The interface is a browser-based single-page style application. Common elements are a persistent navigation bar (Home, About, I'm a Student, I'm a Teacher), a light / dark theme toggle, role dashboards and modal dialogs for focused tasks (create subject, share class link, voice attendance, attendance report, enrol in subject). Detailed screens and design rules are given in Section 10.

6.2 Hardware Interfaces

Table 24: Hardware interfaces

Device	Use
Webcam / phone camera	Capture student face for registration and FaceID login; capture or upload class photographs.
Microphone	Record voice sample at registration and classroom audio for voice attendance.

6.3 Software Interfaces

Table 25: Software interfaces

Interface	Provider	Protocol	Purpose
Database API	Supabase (PostgREST)	HTTPS / REST via supabase-py	Create, read, update and delete application data.
Face models	dlib	In-process Python API	Detection, landmarks and 128-d embeddings.
Classifier	scikit-learn	In-process Python API	SVC identity classification.
Voice encoder	Resemblyzer	In-process Python API	256-d speaker embeddings.
Audio processing	librosa	In-process Python API	Loading, resampling and segmentation.
QR generation	segno	In-process Python API	QR codes for class joining.
Hosting / deploy	Streamlit Community Cloud, Vercel, GitHub	Git push / HTTPS	Build, deploy and serve applications.

6.4 Communication Interfaces

All communication uses HTTPS (TLS). The browser communicates with the Streamlit server over HTTPS and WebSocket as managed by Streamlit. The Streamlit server communicates with Supabase over HTTPS using the project URL and an API key held in the platform's secrets store.

7. System Architecture

7.1 Architectural Overview

Present Call follows a layered architecture that separates presentation, application logic, AI processing and data persistence. The AI engines are called in-process by the application layer, which keeps the system simple to deploy (a single Python application) while preserving clear module boundaries that allow the AI layer to be extracted into a separate service in future.

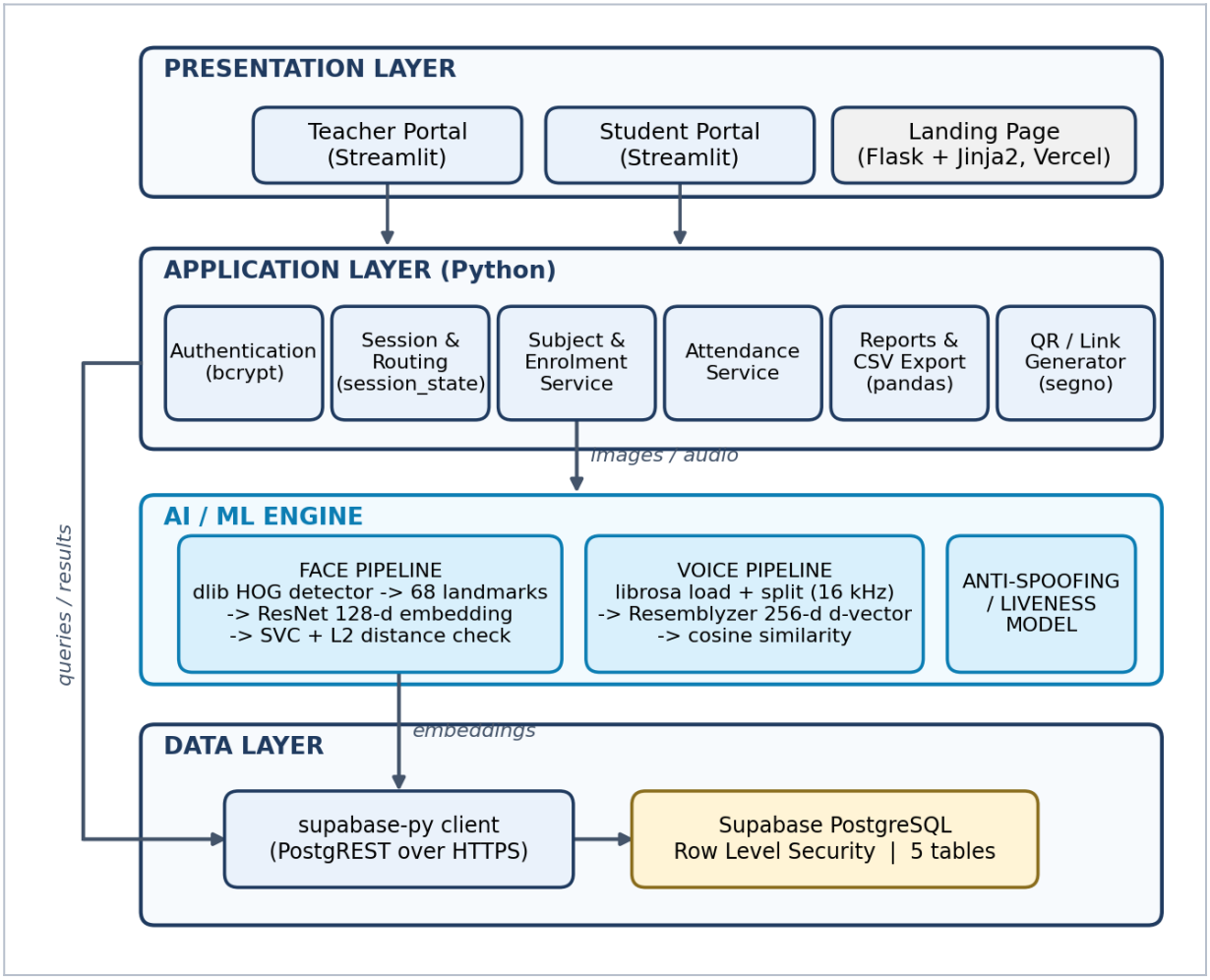


Figure 2: Layered system architecture of Present Call

7.2 Component Descriptions

Table 26: Component responsibilities

Component	Responsibility
Teacher Portal	Teacher login / registration, subject management, take attendance (face and voice), review dialog, records.
Student Portal	FaceID login, enrolment, biometric registration, personal dashboard.
Landing Page (Flask)	Marketing and demonstration site with interactive dashboard simulation; hosted on Vercel and linking to the main application.
Session & Routing	Holds login_type and page in st.session_state and renders the correct view while keeping the navbar and theme persistent.
Authentication	Credential verification with bcrypt; FaceID verification for students.
Subject & Enrolment Service	Creates subjects, validates codes, manages enrolment and unenrolment.
QR / Link Generator	Builds join links and QR images using segno.
Attendance Service	Orchestrates the face and voice pipelines, builds the report and persists confirmed sessions.
Reports & Export	Aggregates records with pandas and produces CSV and trend charts.

Component	Responsibility
Face Pipeline	dlib HOG detector, 68-point shape predictor, ResNet 128-d embedding, SVC and L2 check.
Voice Pipeline	librosa loading and silence-based splitting, Resemblyzer 256-d embedding, cosine similarity.
Anti-Spoofing Model	Liveness decision for faces and voice recordings.
Data Access (supabase-py)	Typed access to Supabase tables via PostgREST.
Supabase PostgreSQL	Persistent storage with indexes and Row Level Security.

7.3 Technology Stack

Table 27: Technology stack

Layer	Technology	Purpose
Application framework	Streamlit (Python)	Frontend and backend logic in one application
Landing page	Flask + Jinja2 on Vercel	Marketing and demonstration site
Database	Supabase (PostgreSQL), supabase-py	Hosted database, REST API, RLS
Face detection	dlib HOG frontal detector	Locate faces
Face landmarks	dlib shape_predictor (68 points)	Align faces
Face embedding	dlib face_recognition_model_v1 (ResNet)	128-d vector per face
Identity matching	scikit-learn SVC (linear) + L2 check	Classify embeddings and reject unknowns
Voice embedding	Resemblyzer VoiceEncoder	256-d d-vector
Audio processing	librosa (load, effects.split)	16 kHz resampling and speech segmentation
Speaker matching	NumPy cosine similarity	Compare live and stored voices
Utilities	Pillow, pandas, NumPy, bcrypt, segno	Images, CSV, maths, password hashing, QR
Theming	Custom CSS via st.markdown; Space Grotesk and Inter fonts	Light / dark design system
Source control	GitHub	Repositories for application and landing page

7.4 Deployment Architecture

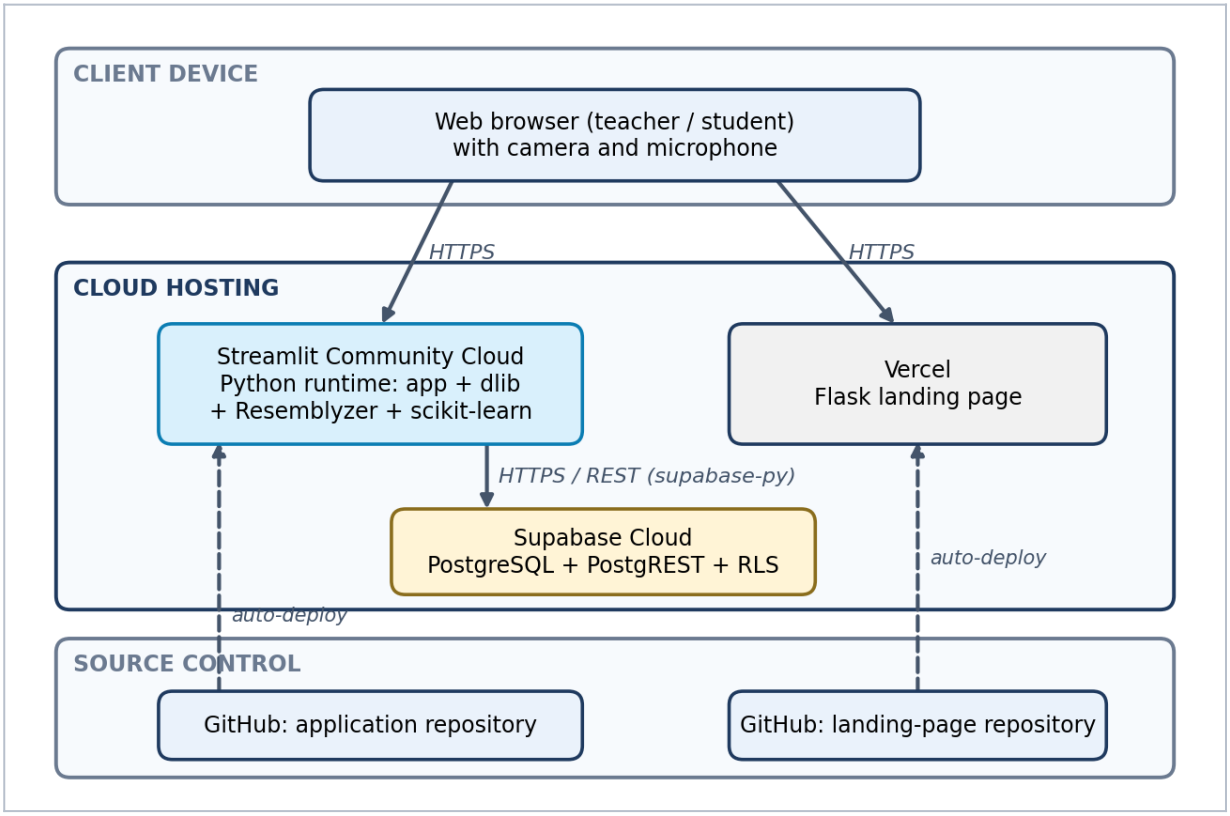


Figure 3: Deployment architecture

7.5 Key Architecture Decisions

Table 28: Architecture decision record

Decision	Rationale	Trade-off
Single Streamlit application	Rapid development in Python; UI and logic in one codebase; easy deployment.	Less UI flexibility; heavy computation shares the web process.
Session-state routing	Shared navbar and theme without page reloads; simple role switching.	Not shareable by URL for every page; requires careful state handling.
dlib embeddings + SVC	Pre-trained, CPU-friendly, strong baseline; SVC adapts to enrolled students.	Retraining needed when roster changes; sensitive to pose and lighting.
Resemblyzer d-vectors	Pre-trained speaker encoder; works without custom training.	Accuracy falls with noise and overlapping speech.
Two-factor biometrics	Independent face and voice modes reduce spoofing risk.	More enrolment effort for students.
Supabase PostgreSQL	Managed SQL, REST API, RLS and real-time features at low cost.	Dependency on a third-party service and plan limits.
Separate Flask landing page	Independent hosting and updates of the public site.	Two repositories to maintain.

8. Detailed Design

8.1 Face Recognition Pipeline

The face pipeline converts each photograph into a set of identified students. Faces are first located, then geometrically normalised, then encoded, and finally classified. The flow is shown in the figure below and the stages are listed in the table that follows it.

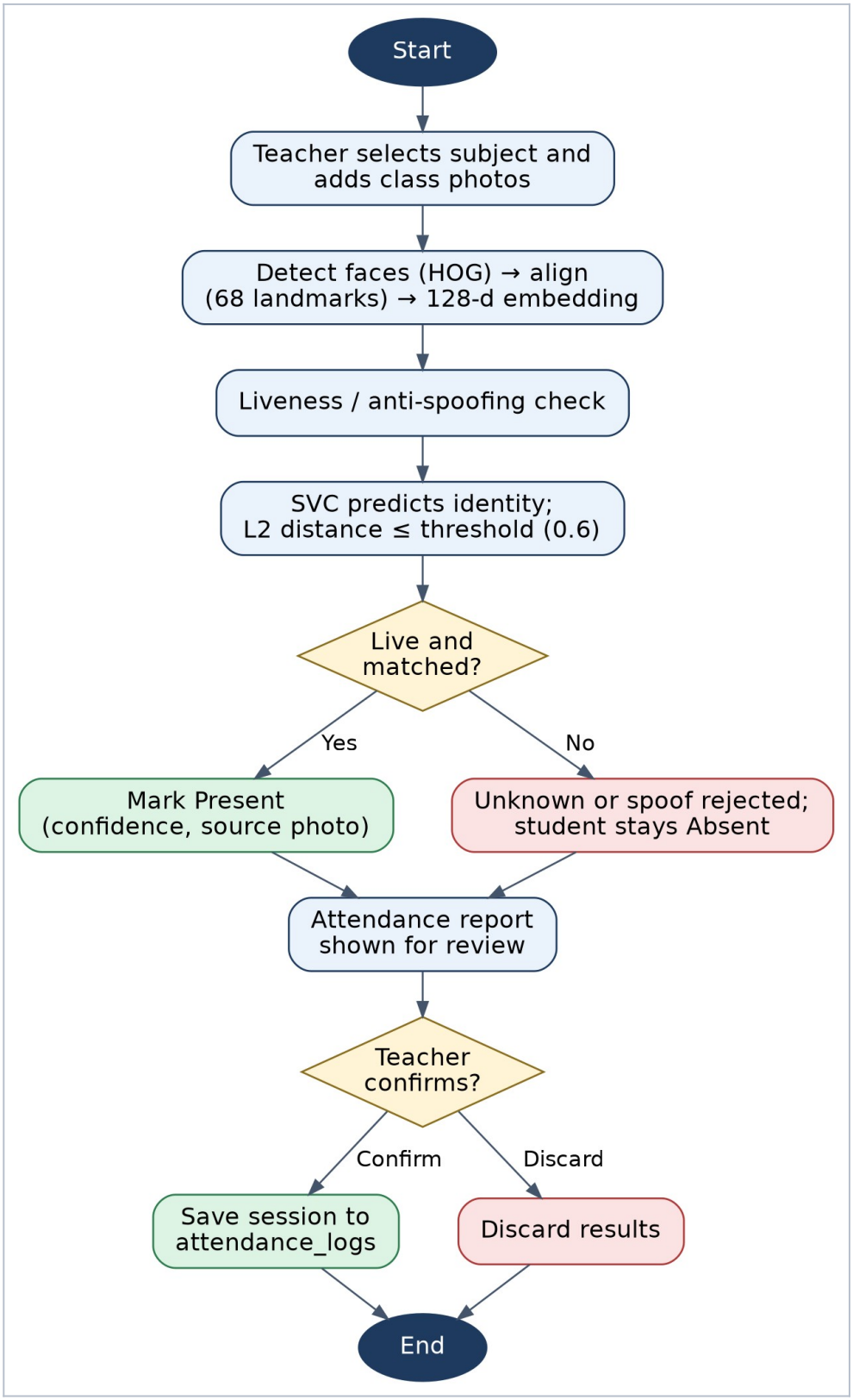


Figure 4: Face attendance flow

Table 29: Face pipeline stages

Step	Stage	Technique	Output
1	Acquisition	Teacher uploads / captures one or more photos	RGB images
2	Detection	dlib HOG-based frontal face detector	Bounding boxes
3	Alignment	dlib 68-point shape predictor	Normalised face chips
4	Embedding	dlib ResNet face_recognition_model_v1	128-d vector per face
5	Liveness	Anti-spoofing model	Live / spoof decision
6	Classification	Linear SVC trained on enrolled embeddings	Predicted student ID
7	Verification	L2 distance to the student's stored embedding vs threshold (0.6 default)	Accept / reject
8	Reporting	Aggregate across photos	Present / Absent per student with source photo

8.2 Voice Recognition Pipeline

The voice pipeline identifies who spoke each "present" in a continuous classroom recording. Silence detection isolates utterances; a pre-trained speaker encoder converts each utterance into a 256-dimensional d-vector which is compared with enrolled voices.

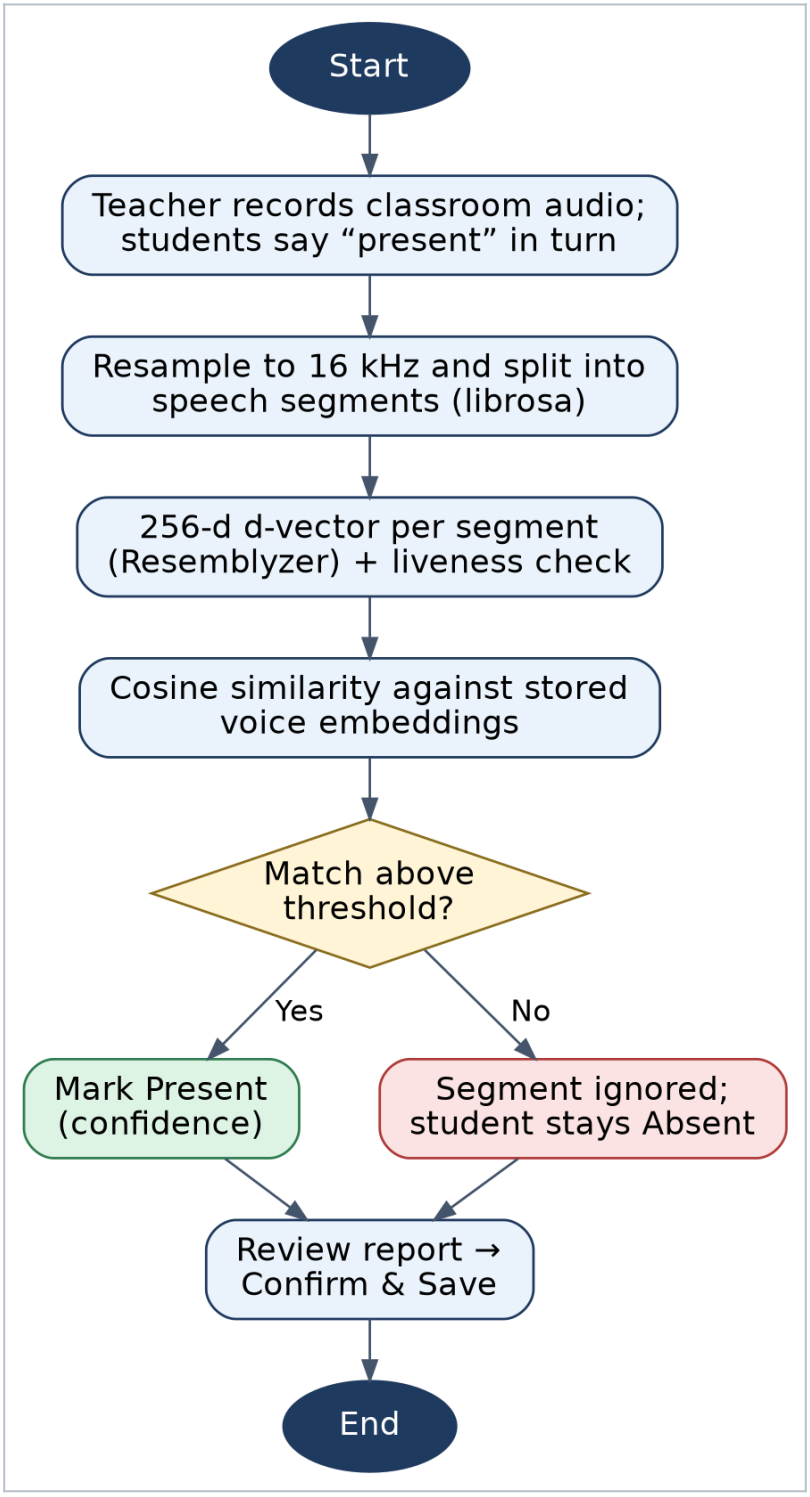


Figure 5: Voice attendance flow

Table 30: Voice pipeline parameters

Parameter	Design value / behaviour
Sample rate	16 kHz (librosa.load resampling)
Segmentation	librosa.effects.split with silence threshold; each segment is one utterance
Encoder	Resemblyzer VoiceEncoder, 256-d d-vector
Similarity	Cosine similarity (dot product of normalised vectors)

Parameter	Design value / behaviour
Decision	Best-matching enrolled student whose similarity exceeds the threshold
Failure handling	Unmatched segments are ignored; student stays Absent

8.3 Decision Logic and Confidence

To reduce false acceptances, a face is accepted only when two independent checks agree: the SVC predicts a student identity, and the Euclidean (L2) distance between the face embedding and that student's stored embedding is within the threshold. Voice uses a similarity threshold on the cosine score. The resulting score is stored as the confidence of the record.

Table 31: Decision rules

Mode	Metric	Rule
Face	L2 distance (lower is better)	Accept if SVC identity matches and distance ≤ threshold (default 0.6).
Voice	Cosine similarity (higher is better)	Accept if best similarity ≥ voice threshold.
Combined (future)	Both modes	Mark present if either mode passes, or require both for high-security sessions.

8.4 Anti-Spoofing and Liveness

A model-based anti-spoofing layer is applied to detected faces and to voice recordings. Its purpose is to reject presentation attacks: printed photographs, photographs displayed on a screen and replayed audio. The module returns a live / spoof decision that must be "live" before an identity match is accepted. Its performance should be evaluated separately from identification accuracy using attack samples (see Section 12.3).

8.5 Biometric Enrolment Design

1. The student opens the Student Portal and captures a face photograph through the webcam (FaceID).
2. The system detects the face, computes the 128-d embedding and stores it against the student record.
3. The student provides a voice sample; the system computes the 256-d embedding and stores it.
4. Only derived embeddings are persisted; raw images and audio are discarded after processing.
5. When enrolment changes the roster for a subject, the SVC is re-fitted on the current set of embeddings before the next attendance run.

8.6 Sequence: Face Attendance Session

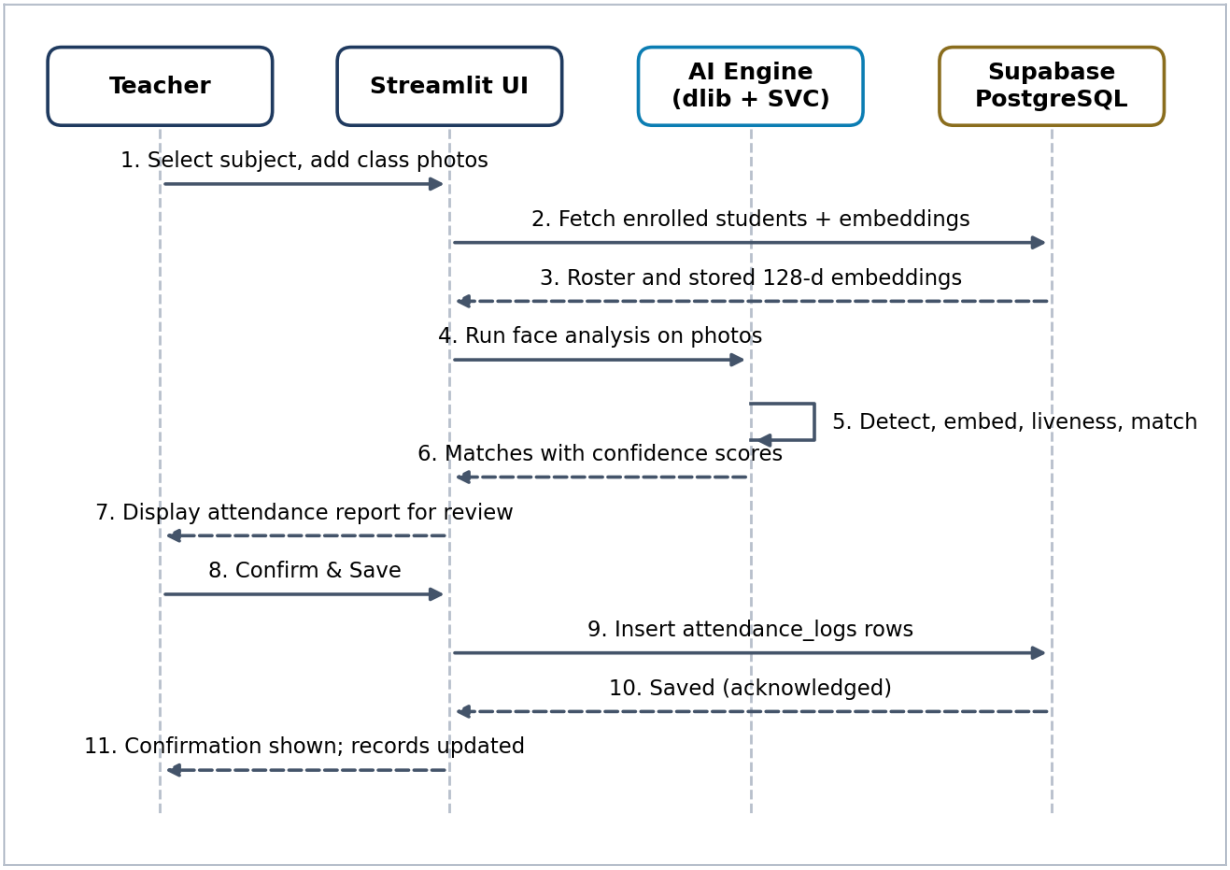


Figure 6: Sequence diagram of a face attendance session

8.7 Error Handling

Table 32: Error handling matrix

Condition	System behaviour	User message / action
No face detected in a photo	Photo skipped; processing continues	Notify teacher; suggest a clearer photo
Unknown face	Not matched; no student marked	Shown only in logs / summary
Spoof detected	Identity match rejected	Flag for teacher review
Silence-only audio	No segments produced	Ask teacher to record again
Microphone / camera blocked	Capture widget unavailable	Show permission instructions
Database unavailable	Save aborted; nothing persisted	Show retry message; keep pending report
Duplicate enrolment	Insert rejected by unique constraint	Inform student already enrolled
Invalid subject code	Lookup returns nothing	Inform student code not found

9. Data Design

9.1 Entity-Relationship Model

The database contains five tables. A teacher owns many subjects; students and subjects are linked many-to-many through subject_students; each attendance record belongs to one student and one subject. A saved session is identified by its subject and session timestamp.

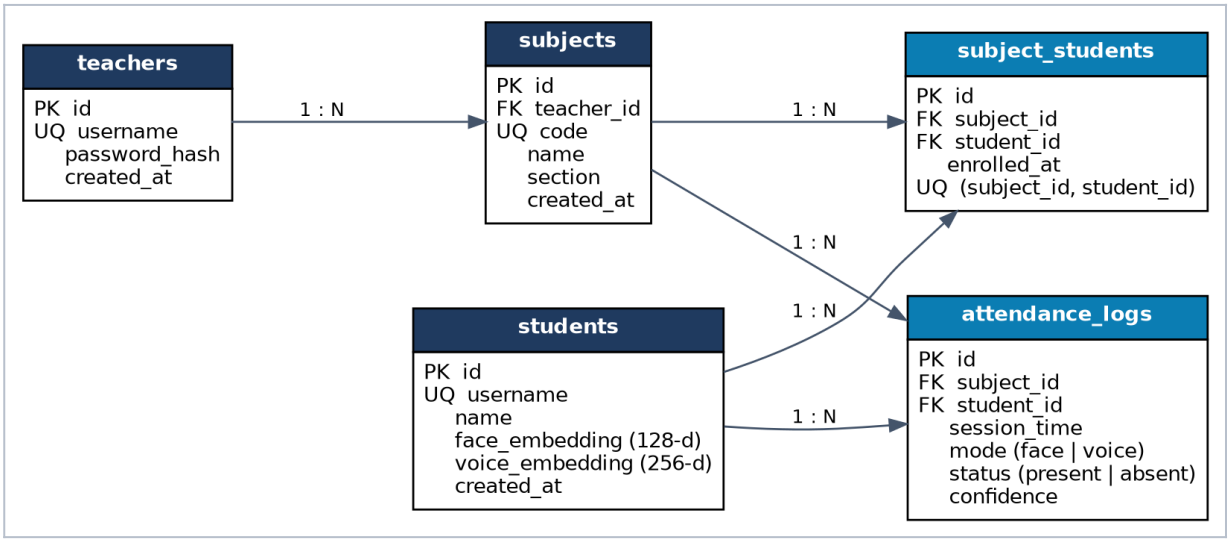


Figure 7: Entity-relationship diagram (logical model)

> **Note:** The column lists below are the logical data model derived from the application behaviour. They must be reconciled with the actual migration scripts in the repository before release.

9.2 Data Dictionary

teachers

Column	Type	Constraint	Description
id	bigint / uuid	Primary key	Unique teacher identifier
username	text	Unique, not null	Login name
password_hash	text	Not null	bcrypt hash of the password
created_at	timestamptz	Default now()	Registration time

students

Column	Type	Constraint	Description
id	bigint / uuid	Primary key	Unique student identifier
username	text	Unique, not null	Student login / display name
name	text	Not null	Full name
face_embedding	float8[128] / jsonb	Not null	128-d face template
voice_embedding	float8[256] / jsonb	Nullable	256-d voice template
created_at	timestamptz	Default now()	Registration time

subjects

Column	Type	Constraint	Description
id	bigint / uuid	Primary key	Unique subject identifier
teacher_id	foreign key	References teachers(id)	Owner of the subject
code	text	Unique, not null	Subject code used to enrol (e.g. CS701)
name	text	Not null	Subject name (e.g. IOT)
section	text	Not null	Section (e.g. A)
created_at	timestamptz	Default now()	Creation time

subject_students

Column	Type	Constraint	Description
id	bigint / uuid	Primary key	Row identifier
subject_id	foreign key	References subjects(id)	Subject
student_id	foreign key	References students(id)	Student
enrolled_at	timestamptz	Default now()	Enrolment time
(subject_id, student_id)	composite	Unique	Prevents duplicate enrolment

attendance_logs

Column	Type	Constraint	Description
id	bigint / uuid	Primary key	Row identifier
subject_id	foreign key	References subjects(id)	Subject of the session
student_id	foreign key	References students(id)	Student
session_time	timestamptz	Not null, indexed	Date and time of the class session
mode	text	face | voice	Method used to mark attendance
status	text	present | absent	Attendance result
confidence	float8	Nullable	Match confidence score

9.3 Indexing and Integrity

- Unique indexes on teachers.username, students.username and subjects.code.
- Composite unique index on subject_students (subject_id, student_id).
- Index on attendance_logs (subject_id, session_time) for session listing and (student_id, subject_id) for dashboards.
- Foreign keys with cascading rules defined so that deleting a subject removes its enrolments and logs.

9.4 Data Volume Estimate

A face embedding of 128 double-precision values occupies about 1 KB and a voice embedding of 256 single-precision values about 1 KB, so 1,000 enrolled students need roughly 2 MB of template data. For attendance logs, a campus with 500 students attending 5 lectures a day over 200 teaching days produces about 500,000 rows a year; at roughly 100 bytes per row this is about 50 MB a year, comfortably within small cloud-database plans. These figures are estimates for planning only.

9.5 Data Retention

Table 33: Retention rules

Data	Retention rule (recommended)
Raw photos and audio	Not stored; discarded immediately after embedding extraction.
Biometric templates	Kept while the student is active; deleted on request or when the student leaves the institution.
Attendance logs	Kept for the period required by institutional policy (typically the academic programme plus audit period).
Application logs	Short rotation (for example 30–90 days); must not contain biometric data.

10. User Interface and Experience Design

10.1 Design System

Table 34: Design system

Element	Specification
Typography	Space Grotesk for headings; Inter for body text, loaded through Google Fonts with system fallbacks.
Colour palette	Restricted palette of black, grey, white and sky blue; sky blue is the primary action colour; defined as a THEMES dictionary.
Themes	Light and dark themes toggled from the navigation bar and stored in st.session_state.
Layout	Centred content container with a persistent navbar (Home, About, I’m a Student, I’m a Teacher), card-based lists and modal dialogs.
Components	Custom-built navbar, footer, subject cards, statistic chips, buttons with keyboard-shortcut hints and data tables.
Interaction	Primary action in sky blue, secondary actions in black or white; Ctrl+Enter to log in, Ctrl+Backspace to log out.
Feedback	Status icons (present / absent), confidence values and confirm / discard dialogs for destructive or persisting actions.

10.2 Screen Inventory

Table 35: Screen inventory

Screen	Role	Purpose and key elements
Teacher Login	Teacher	Username and password fields, Login (Ctrl+Enter) and Register Instead buttons.
Take Attendance	Teacher	Subject selector, Add Photos, Clear all photos, Run Face Analysis, Use Voice Attendance.
Manage Subjects	Teacher	Subject cards with code, section, student and class counts; New Subject; Share Code.
Create New Subject	Teacher	Modal with subject code, name and section.
Share Class Link	Teacher	Modal with copyable link, subject code and QR code.
Voice Attendance	Teacher	Modal with audio recorder and Analyze Audio.
Attendance Report	Teacher	Modal table of name, ID, source and status; Discard and Confirm & Save.
Attendance Records	Teacher	Table of sessions: time, subject, subject code, attendance statistics.
Student FaceID Login	Student	Webcam preview and Take Photo.
Student Dashboard	Student	Enrolled subject cards with total and attended classes; Enroll in Subject; Unenroll.
Enroll in Subject	Student	Modal asking for the subject code provided by the teacher.

10.3 Teacher Interface

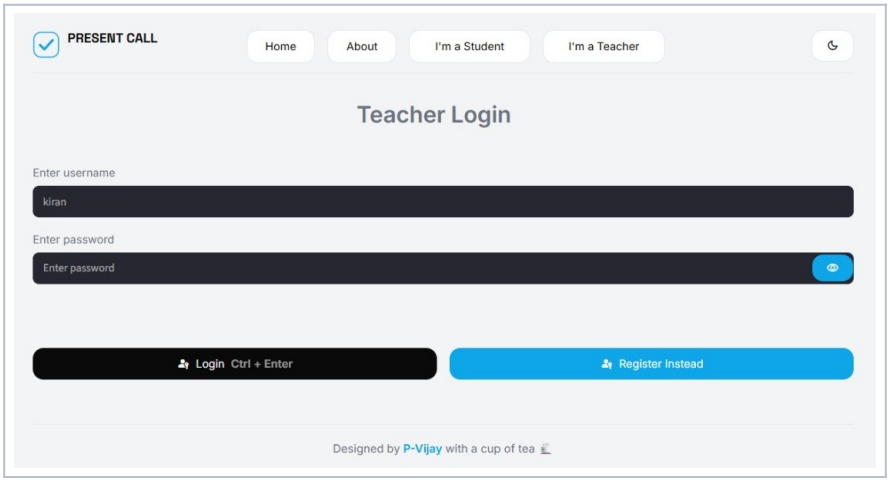


Figure 8: Teacher login screen

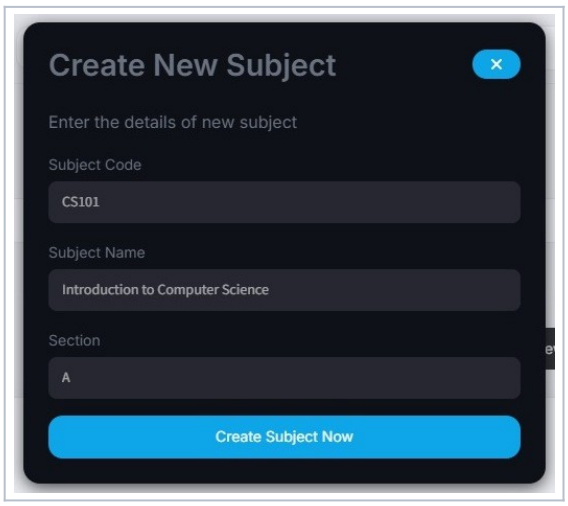


Figure 9: Create New Subject dialog



Figure 10: Share Class Link with QR code

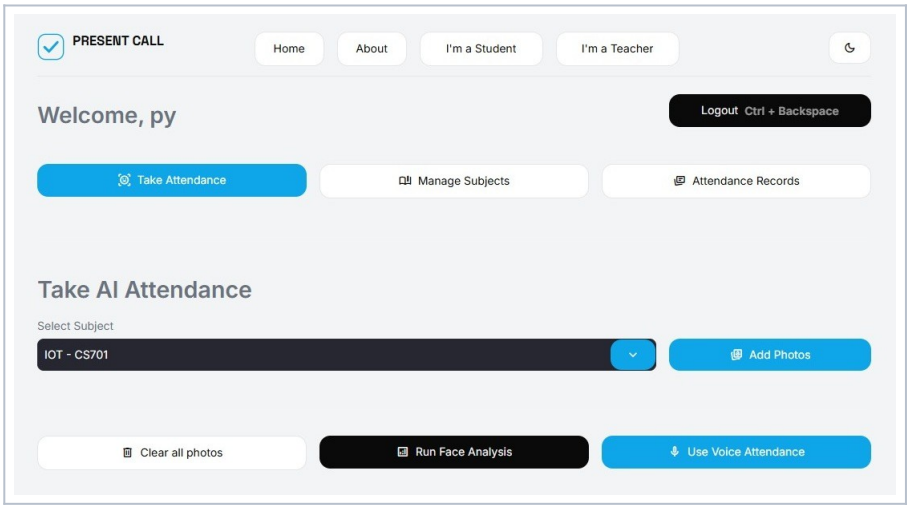


Figure 11: Take AI Attendance screen

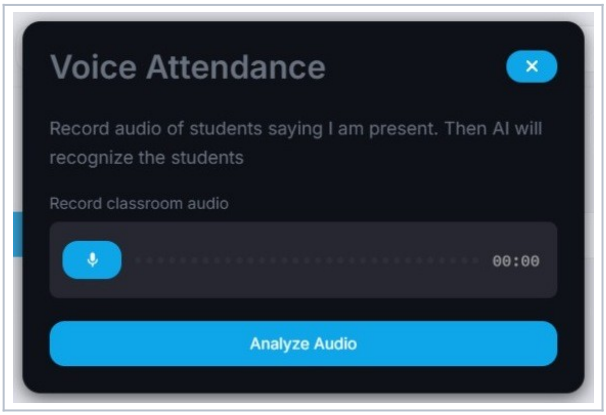


Figure 12: Voice Attendance dialog

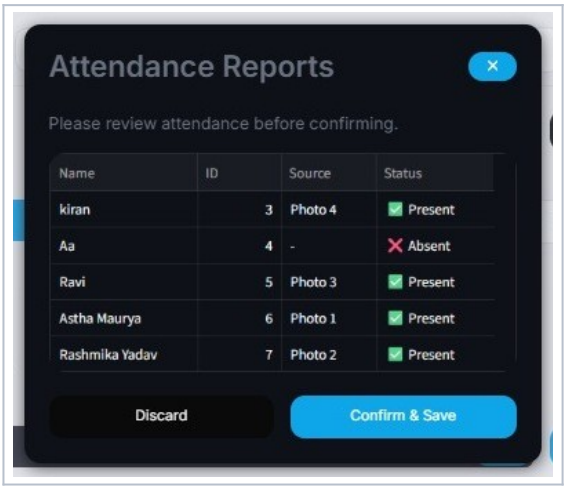


Figure 13: Attendance report for review

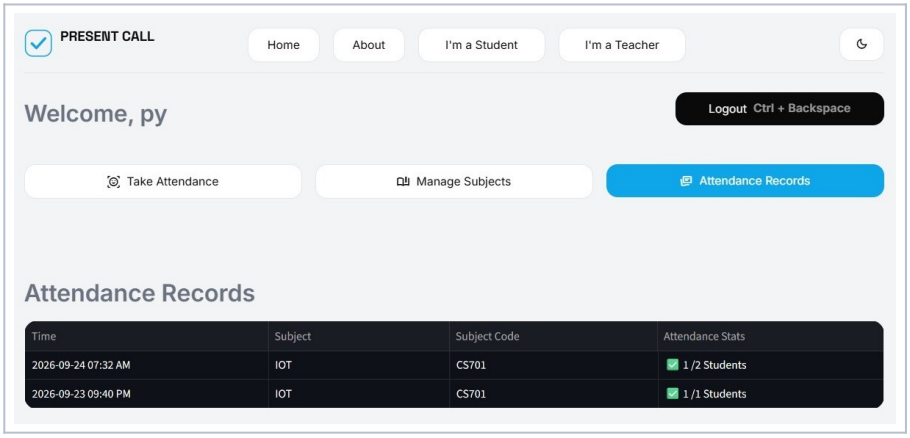


Figure 14: Attendance Records

10.4 Student Interface

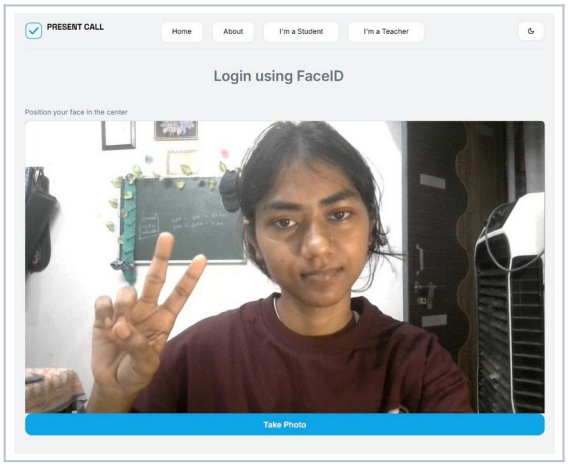


Figure 15: Student FaceID login

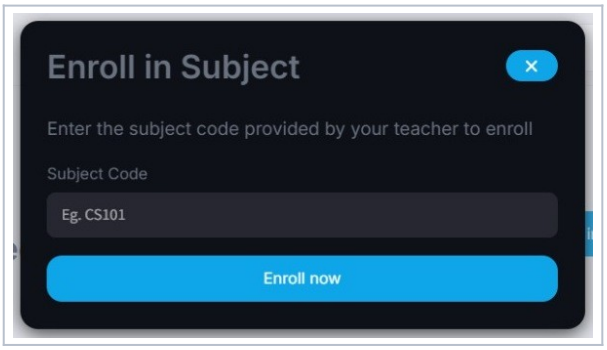


Figure 16: Enroll in Subject dialog

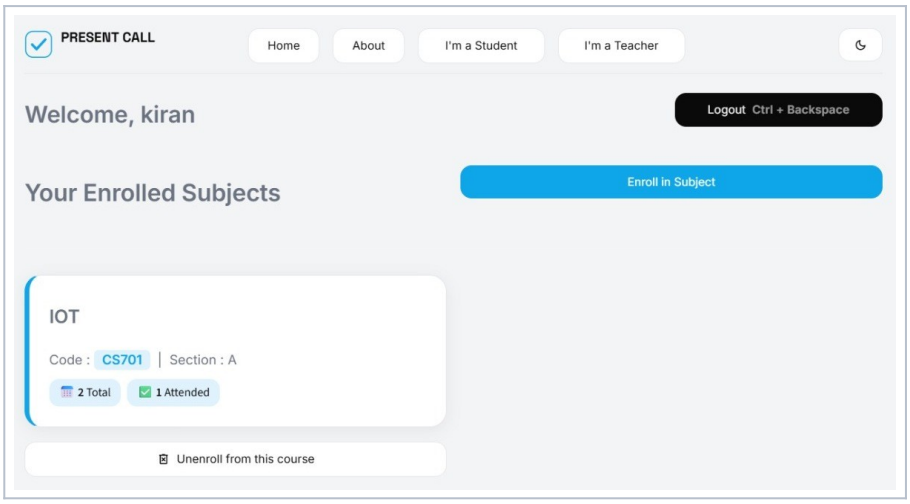


Figure 17: Student dashboard

10.5 Usability and Accessibility Guidelines

- Keep the teacher’s primary flow to a few steps: log in, choose subject, add photos, analyse, review, save.
- Use plain labels and icons together; confirm before persisting data.
- Maintain sufficient colour contrast in both themes and visible focus states for keyboard users.

- Provide clear guidance when camera or microphone permission is denied.
- Design for mobile widths so students can enrol and register from phones.

11. Security and Privacy Design

11.1 Security Objectives

The system must protect the confidentiality and integrity of accounts and biometric templates, guarantee that attendance records cannot be tampered with by unauthorised users, and resist presentation (spoofing) attacks against the biometric modes.

11.2 Threat Analysis

Table 36: Threat analysis

Threat	Vector	Mitigation	Residual risk
Photo / screen spoofing	Printed photo or phone screen shown to camera	Anti-spoofing model; teacher review before save	Medium
Voice replay	Pre-recorded “present”	Liveness check on audio; teacher supervision	Medium
Credential attacks	Guessing or stuffing teacher passwords	bcrypt hashing; planned rate limiting and strong password policy	Medium
Unauthorised data access	Direct calls to database API with exposed key	Production RLS policies; server-side secrets; least-privilege keys	High until RLS hardened
Injection	Malicious input in forms	Parameterised queries through PostgREST; input validation	Low
Template theft	Database leak exposing embeddings	Access control; encryption at rest by provider; minimal data	Medium
Man-in-the-middle	Interception of traffic	HTTPS / TLS everywhere	Low
Cross-user access (IDOR)	Student altering identifiers to view others	Authorisation checks bound to the logged-in user	Medium
Secret leakage	Keys committed to GitHub	Platform secrets; .gitignore; key rotation	Low
Service abuse / DoS	Excessive uploads or requests	Upload size limits; platform rate limits	Medium

11.3 Authentication and Authorisation

- Teachers authenticate with username and bcrypt-verified password; students authenticate with FaceID.
- The role (login_type) in session state controls which pages are rendered; every data query is scoped to the logged-in user’s identifier.
- Teachers may only manage subjects they own; students may only see their own enrolments and attendance.
- Recommended enhancement: move to Supabase Auth or signed tokens so that database policies can use authenticated identities.

11.4 Biometric Data Protection

- Treat face and voice embeddings as sensitive personal data even though they are not raw images; they can still identify a person.
- Store only embeddings; discard raw images and audio after processing.
- Restrict embedding columns to the application's server role; never expose them to client-side queries.
- Do not write biometric data or images to logs.

11.5 Privacy and Legal Compliance

Biometric attendance involves personal data of students, many of whom may be minors in some institutions. The deploying institution should adopt the privacy controls below and review them with legal counsel under the Digital Personal Data Protection Act, 2023 and applicable institutional policy. The recommended controls are listed below.

Table 37: Privacy controls

Principle	Control
Consent and notice	Show a clear consent screen before biometric registration stating purpose, data stored and retention period.
Purpose limitation	Use biometric data only for attendance; no secondary use or sharing without consent.
Data minimisation	Store embeddings only; no raw images; no unnecessary personal fields.
Accuracy and correction	Allow re-enrolment and correction of templates.
Retention and deletion	Delete templates on request or when no longer needed; document retention periods.
Security safeguards	Access control, encryption in transit and at rest, key management, incident response.
Alternative method	Provide a non-biometric fallback (manual marking by teacher) for students who do not consent.

11.6 Production Hardening Checklist

> **Important:** The Row Level Security policies in the current development setup are permissive. They must be replaced with restrictive policies before the system is used with real student data.

Table 38: Hardening checklist

#	Action	Status
1	Replace permissive RLS policies with per-role, least-privilege policies on all five tables.	Open
2	Use a restricted key in the application; keep any service-role key out of the client path.	Open
3	Add rate limiting and lock-out for repeated failed teacher logins.	Open
4	Add consent capture and a documented privacy notice.	Open
5	Verify that no raw images or audio are persisted anywhere (storage, logs, cache).	Open

#	Action	Status
6	Rotate API keys and confirm secrets are stored only in the platform secrets manager.	Open
7	Enable database backups and test restoration.	Open
8	Run dependency vulnerability scan and fix high-severity findings.	Open
9	Add upload size and type limits for photos and audio.	Open
10	Perform a security review against OWASP ASVS Level 1.	Open

12. Testing and Quality Assurance

12.1 Test Strategy

Table 39: Test levels

Level	Scope	Approach
Unit testing	Embedding helpers, distance and similarity functions, bcrypt utilities, data-access functions	pytest with mocked database and fixture images / audio
Integration testing	Pipelines with the database; enrolment and attendance flows	Test Supabase project with seeded data
System testing	End-to-end scenarios through the Streamlit UI	Manual scripted tests; optional browser automation
Biometric evaluation	Accuracy, FAR, FRR, spoof rejection	Protocol in Section 12.3
Security testing	Authentication, authorisation, injection, secrets	Checklist from Section 11.6; dependency scan
Usability testing	Teacher and student task completion	Observed sessions with a small user group
Acceptance testing	Business objectives and critical requirements	Criteria in Section 12.6

12.2 Test Environment

- Local environment: Python virtual environment, a test Supabase project, a webcam and microphone.
- Cloud environment: staging deployment of the Streamlit application against a non-production database.
- Test data: synthetic or consenting-volunteer face and voice samples; no real student data in testing.

12.3 Biometric Evaluation Methodology

Biometric quality must be reported with metrics that are independent of the number of users. The following definitions are used.

Table 40: Biometric metrics

Metric	Definition
Accuracy	Correct identifications ÷ total identification attempts.
FAR (False Acceptance Rate)	Impostor or unknown attempts wrongly accepted ÷ total impostor attempts.
FRR (False Rejection Rate)	Genuine attempts wrongly rejected ÷ total genuine attempts.
EER (Equal Error Rate)	Operating point at which FAR equals FRR; used to compare configurations.
Spoof rejection rate	Presentation attacks (photo, screen, replay) correctly rejected ÷ total attacks.

Protocol. Enrol a set of volunteers; collect test captures under varied lighting, distance, pose, accessories and background noise; include unknown people as impostors; include printed-photo, screen-replay and audio-replay attacks; record results in a confusion matrix; report each metric with the number of samples and a confidence interval.

12.4 Reported Result

Observed accuracy	95.8% identification accuracy in internal testing.
Models	Pre-trained dlib face models and Resemblyzer voice encoder; SVC fitted on enrolled students' embeddings.
Dataset	No dedicated public benchmark dataset was used; testing used enrolled project users.
Recommendation	Repeat the evaluation using the protocol above with a larger, documented sample and report FAR, FRR and spoof rejection in addition to accuracy.

12.5 Test Case Catalogue

Table 41: Test case catalogue

ID	Area	Scenario	Expected result	Req.
TC-01	Auth	Register teacher with new username	Account created; password stored hashed	FR-AUT-01/02
TC-02	Auth	Login with wrong password	Login rejected with error	FR-AUT-03
TC-03	Auth	Student FaceID login with registered face	Dashboard opens	FR-AUT-04
TC-04	Subject	Create subject with unique code	Subject card shown	FR-SUB-01/02
TC-05	Subject	Create subject with duplicate code	Creation rejected	FR-SUB-01
TC-06	Enrol	Student enrols with valid code	Subject appears on dashboard	FR-ENR-01
TC-07	Enrol	Enrol twice in same subject	Duplicate prevented	FR-ENR-02

ID	Area	Scenario	Expected result	Req.
TC-08	Enrol	Enrol with invalid code	Error: code not found	FR-ENR-01
TC-09	Face	Group photo with 4 enrolled students	All 4 marked present with source photo	FR-FAC-02/05
TC-10	Face	Photo with enrolled student absent	Absent student marked absent	FR-FAC-04
TC-11	Face	Photo with unknown person	Unknown not matched to any student	FR-FAC-08
TC-12	Face	Printed photograph of a student	Rejected as spoof	FR-FAC-06
TC-13	Face	Photo without faces	Handled gracefully; no crash	FR-FAC-02
TC-14	Voice	Students say “present” in turn	Matching students marked present	FR-VOI-01…04
TC-15	Voice	Recorded playback of a voice	Rejected as replay	FR-VOI-05
TC-16	Voice	Silent recording	No segments; prompt to re-record	FR-VOI-02
TC-17	Review	Discard report	No data persisted	FR-REV-03
TC-18	Review	Confirm & Save	Session saved; counts update in records	FR-REV-02, FR-REC-01
TC-19	Reports	Export CSV	File contains all selected records	FR-REC-02
TC-20	Security	Student tries to access another student’s data	Access denied	NFR-SEC-05
TC-21	UI	Toggle light / dark theme	Theme persists across pages	FR-ADM-04
TC-22	Perf	Analyse photo with 30 faces	Completes within target time	NFR-PER-01

12.6 Acceptance Criteria

- All “Must” functional requirements marked Implemented pass their test cases.
- Identification accuracy is at least 95% on the agreed evaluation set.
- All planned spoof attacks (printed photo, screen photo, audio replay) are rejected in testing.
- No critical or high-severity defects remain open.
- Production hardening checklist items 1, 4, 5 and 6 are closed before real student data is used.

12.7 Defect Severity Classification

Table 42: Defect severity

Severity	Definition	Response
Critical	Data loss, security breach, system unusable	Fix immediately; block release
High	Major feature fails or wrong attendance recorded	Fix before release
Medium	Feature works with a workaround	Schedule for next release
Low	Cosmetic or minor inconvenience	Backlog

13. Deployment and Operations

13.1 Environments

Table 43: Environments

Environment	Description	Notes
Development	Developer laptop with virtual environment	Test Supabase project; local Streamlit server
Staging (optional)	Separate Streamlit app and database	Pre-release verification
Production	Streamlit Community Cloud + Supabase + Vercel	Deployed from the main branch on GitHub

13.2 Build and Release Pipeline

1. Developer commits changes to a feature branch and opens a pull request on GitHub.
2. Automated checks run: formatting / linting, unit tests, dependency scan (recommended).
3. After review, the branch is merged into main.
4. Streamlit Community Cloud and Vercel automatically redeploy from main.
5. The release is tagged using semantic versioning (MAJOR.MINOR.PATCH) and noted in the changelog.

13.3 Configuration and Secrets

Table 44: Configuration items

Setting	Purpose	Where stored
Supabase project URL	Database endpoint	Platform secrets
Supabase API key	Authenticates database calls	Platform secrets (never in Git)
Face distance threshold	Acceptance threshold (default 0.6)	Application configuration
Voice similarity threshold	Acceptance threshold for speaker match	Application configuration
Audio sample rate	16 kHz	Application constant
Theme palette	Light / dark colours	THEMES dictionary

13.4 Local Installation Guide

The steps below set up a development machine. Replace the placeholder names with the actual repository and entry file.

1. **Prerequisites:** Python 3.x, Git, a C++ build toolchain and CMake (required to build dlib).
2. **Clone:** `git clone <repository-url>` and change into the folder.
3. **Virtual environment:** `python -m venv venv` then activate it.
4. **Install dependencies:** `pip install -r requirements.txt` (includes streamlit, supabase, dlib, scikit-learn, resemblyzer, librosa, pandas, numpy, pillow, bcrypt, segno).
5. **Configure secrets:** create the Streamlit secrets file with the Supabase URL and key.
6. **Initialise database:** run the SQL migration to create the five tables, indexes and policies.
7. **Run:** `streamlit run <entry_file>.py` and open the displayed local URL.

13.5 Monitoring and Logging

- Application logs from the hosting platform for errors and slow requests (without biometric data).
- Supabase dashboard for database health, API usage and storage.
- Simple uptime check of the public URLs.
- Track key metrics: sessions per day, analysis time, error counts.

13.6 Backup and Disaster Recovery

Table 45: Backup and recovery

Item	Plan
Backups	Enable automatic Supabase backups; additionally export attendance tables periodically as CSV.
Recovery objectives	Target RPO 24 hours and RTO 4 hours.
Restore test	Restore into a staging project at least once per term.
Source code	Protected in GitHub with branch protection on main.

13.7 Operational Runbook

Table 46: Runbook

Incident	Likely cause	Action
App not loading or very slow after idle	Free-tier app asleep / cold start	Reload; consider keep-alive or upgrade plan
“No face detected” for clear photo	Low light, small faces, extreme angle	Retake photo closer and better lit
Many students wrongly absent	Poor enrolment images; threshold too strict	Re-enrol; review threshold; evaluate FRR
Database errors on save	Network issue, key expired, policy block	Check Supabase status, key and policies
Microphone not working	Browser permission denied	Enable microphone permission for the site
Build fails on deploy	dlib or dependency version mismatch	Pin versions; check build logs

14. Project Management

14.1 Team and Governance

Project	Present Call: AI-Based Attendance System using Face and Voice Biometrics
Team	Priyanshu Vijay (238330082), Ravi Kumar Gangwar (238330087), Soumya Pandey (238330116)
Guide	Adarsh Vardhan, Assistant Professor, Department of Computer Science and Engineering
Institution	Dr. Shakuntala Misra National Rehabilitation University — B.Tech CSE, Session 2023–2027

14.2 Development Methodology

The project follows an iterative, incremental approach: requirements are grouped into modules, each module is designed, implemented, tested and demonstrated to the guide before the next, and feedback is folded into the following iteration. A product backlog ordered by MoSCoW priority drives the work.

14.3 Project Phases and Deliverables

Table 47: Phases and deliverables

Phase	Activities	Deliverables
1. Planning	Problem study, literature survey, feasibility, tool selection	Synopsis, plan
2. Requirements	Stakeholder needs, use cases, requirement prioritisation	This SRDS (Sections 1–5)
3. Design	Architecture, AI pipelines, data model, UI design	Design diagrams, schema, mock screens
4. Implementation	Auth, subjects, enrolment, face and voice attendance, records, dashboards	Working application
5. Testing	Unit, integration, biometric evaluation, security checks	Test results, defect log
6. Deployment	Cloud deployment, landing page, configuration	Live application and website
7. Review and handover	Demonstration, documentation, feedback	Final report and presentation

14.4 Risk Register

Table 48: Risk register

ID	Risk	Prob.	Impact	Mitigation
R-01	Accuracy drops in poor light, small faces or extreme pose	High	High	Guidelines for photo capture; multiple photos; re-enrolment; tune threshold
R-02	Spoofing with photos or recordings	Medium	High	Anti-spoofing model; teacher review; periodic attack testing

ID	Risk	Prob.	Impact	Mitigation
R-03	Privacy / legal non-compliance for biometric data	Medium	High	Consent, minimisation, retention policy, legal review
R-04	Permissive database policies exposing data	High	High	Complete hardening checklist before real use
R-05	Free-tier limits cause slowness or downtime	High	Medium	Optimise image size; upgrade plan; cache models
R-06	dlib build / dependency issues on deployment	Medium	Medium	Pin versions; document build prerequisites
R-07	Browser blocks camera / microphone	Medium	Medium	Clear permission guidance; fallback manual entry
R-08	Students refuse biometric consent	Medium	Medium	Provide a non-biometric alternative
R-09	Third-party service change or outage	Low	High	Backups; keep portable architecture

14.5 Cost and Resources

The solution is built entirely on open-source libraries and free-tier cloud services, so direct software and hosting cost for the prototype is negligible. Costs arise at scale from paid hosting plans (more CPU and memory for the AI pipelines), database plan upgrades, backups and any commercial support or security assessment required by an institution.

15. Maintenance, Limitations and Roadmap

15.1 Maintenance Plan

Table 49: Maintenance types

Type	Activities
Corrective	Fix defects reported by users; hot-fix critical issues.
Adaptive	Update dependencies, Python version and platform APIs; respond to provider changes.
Perfective	Improve accuracy, speed and usability based on feedback and measured metrics.
Preventive	Refactor, add tests, rotate secrets, review security regularly.

15.2 Known Limitations

- Accuracy depends on lighting, camera quality, face size and pose, and on audio noise for voice mode.
- Free-tier hosting may be slow after inactivity and limits concurrent heavy processing.
- Database policies are permissive in the development configuration and must be hardened.
- Guardian alerts, PDF reports, attendance percentage and minimum-attendance warnings are planned but not yet part of v1.0.
- Identification accuracy has been measured on a small internal group; larger-scale validation is pending.

15.3 Product Roadmap

Table 50: Roadmap

Release	Theme	Planned features
v1.0 (current)	Core platform	Auth, subjects, QR enrolment, face and voice attendance, anti-spoofing, review, records, CSV export, student dashboard, light / dark theme
v1.1	Compliance and insight	Consent screen, hardened RLS, attendance percentage, minimum-attendance alerts, PDF reports, manual override, subject edit / delete
v1.2	Experience	Installable web app (PWA) for mobile, rate limiting, password reset, configurable thresholds
v2.0	Institution scale	Administrator role, guardian SMS / e-mail alerts, ERP / LMS integration, analytics and prediction, classroom-camera automation

15.4 Conclusion

Present Call combines mature, open-source computer-vision and speaker-recognition models with a simple web workflow to make attendance fast, transparent and harder to forge. This specification records the requirements, design, quality plan and operating procedures needed to take the project from an academic prototype towards a dependable, privacy-respecting institutional tool. The first priorities for that journey are hardening data access, capturing consent, and validating accuracy and spoof resistance on a larger, documented sample.

Appendix A: Requirements Traceability Matrix

Table 51: Traceability matrix

Requirement group	Use case	Test cases	Design section
FR-AUT (authentication)	UC-01, UC-04	TC-01, TC-02, TC-03	11.3
FR-SUB (subjects)	UC-02, UC-03	TC-04, TC-05	9.2
FR-ENR (enrolment)	UC-05, UC-11	TC-06, TC-07, TC-08	8.5
FR-BIO (biometric registration)	UC-04	TC-03	8.5
FR-FAC (face attendance)	UC-06	TC-09 to TC-13	8.1, 8.3, 8.4
FR-VOI (voice attendance)	UC-07	TC-14 to TC-16	8.2, 8.3, 8.4
FR-REV (review)	UC-08	TC-17, TC-18	8.6
FR-REC (records)	UC-09	TC-18, TC-19	9.2
FR-DSH (student dashboard)	UC-10	TC-06	10.4
FR-ADM (administration)	—	TC-21	10.1
NFR-PER (performance)	—	TC-22	12.3

Requirement group	Use case	Test cases	Design section
NFR-SEC (security)	—	TC-12, TC-15, TC-20	11

Appendix B: Key Algorithm Parameters

Table 52: Algorithm parameters

Parameter	Value	Notes
Face embedding dimension	128	dlib ResNet model
Face landmarks	68 points	dlib shape predictor
Face detector	HOG frontal	dlib
Face classifier	SVC, linear kernel	scikit-learn
Face distance threshold	0.6 (default)	Lower is stricter
Voice embedding dimension	256	Resemblyzer d-vector
Audio sample rate	16 kHz	librosa.load
Voice segmentation	librosa.effects.split	Silence-based
Voice similarity	Cosine	numpy dot product
Password hashing	bcrypt	Per-user salt

Appendix C: Sample CSV Export Format

The exported attendance file is a UTF-8 CSV with one row per student per session. A recommended column layout is shown below.

Table 53: Sample CSV rows (illustrative values)

session_time	subject_code	student	status	mode	confidence
2026-09-24 07:32	CS701	Student A	present	face	0.96
2026-09-24 07:32	CS701	Student B	absent	face	—
2026-09-23 21:40	CS701	Student A	present	voice	0.91